



Final exam

Exercise 1.

Let the set $E = \{a, b, c\}$. We denote by $\text{Card}(E)$ the number of elements of E .

1. Determine the power set $\mathcal{P}(E)$ of E .

2. Consider the relation $\mathcal{R}$ defined on $\mathcal{P}(E)$ by

$$\forall A, B \in \mathcal{P}(E), A \mathcal{R} B \iff \text{Card}(A) = \text{Card}(B)$$

(a) Show that $\mathcal{R}$ is an equivalence relation.

(b) Determine all the equivalence classes of A for each $A \in \mathcal{P}(E)$.

Exercise 2.

Let f be an application defined by

$$f : \mathbb{R} \longrightarrow \mathbb{R} \\ x \longmapsto \frac{x}{\sqrt{x^2+1}}$$

1. Is f injective? surjective?

2. Is f bijective? If yes, determine its inverse. If not, give the largest sets E (domain) and F (codomain) such that f is bijective from E to F , then find f^{-1} .

Exercise 3.

Let the complex number $z = \frac{1+i}{\sqrt{2}}$

1. Write z in exponential and trigonometric form.

2. Calculate the square roots of z in two ways and deduce the values of $\cos(\frac{\pi}{8})$ and $\sin(\frac{\pi}{8})$.

3. Find the fourth roots of z .

Exercise 4.

Let $\mathcal{F}(\mathbb{R}, \mathbb{R})$ be the vector space of all real-valued functions defined on $\mathbb{R}$.

Let F be the set of even functions and G the set of odd functions, defined by $F = \{f \in \mathcal{F}(\mathbb{R}, \mathbb{R}) \mid \forall x \in \mathbb{R}, f(-x) = f(x)\}$, $G = \{f \in \mathcal{F}(\mathbb{R}, \mathbb{R}) \mid \forall x \in \mathbb{R}, f(-x) = -f(x)\}$.



Resit exam

Exercise 1.

- Let F and G and two subsets of the set E . Show that $C_E^{(F \cap G)} = C_E^F \cup C_E^G$.
- We define the following relation on the set $\mathbb{R}$:

$$x \mathcal{R} y \Leftrightarrow |x^2 - 1| = |y^2 - 1|.$$

1. Show that $\mathcal{R}$ is an equivalence relation.
2. Determine the equivalence class of 0 and 1.

Exercise 2.

Let f be an application defined by

$$f : \mathbb{R} - \{\ln(2)\} \longrightarrow \mathbb{R} \\ x \longmapsto \frac{1+3e^x}{2-e^x}$$

1. Is f injective? surjective?
2. Is f bijective? If yes, determine its inverse. If not, give the largest set F such that f is bijective from $\mathbb{R}$ to F , then find f^{-1} .

Exercise 3.

Let $U = \frac{\sqrt{3}+i}{1+i\sqrt{3}}$ a complex number.

1. Express U in algebraic and exponential form
2. Calculate the square roots of U in two ways and deduce the values of $\cos(-\frac{\pi}{12})$ and $\sin(-\frac{\pi}{12})$.
3. Solve in $\mathbb{C}$ the equation : $Z^5 = U$.

Exercise 4.

Let $E = \mathbb{R}^4$. We consider the following subsets of E :

$$F = \{(x, y, z, t) \in \mathbb{R}^4 \mid x + y + z + t = 0\}$$

and

$$G = \{(2a, -a, 0, a) \mid a \in \mathbb{R}\}.$$

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1. Show that F and G are vector subspaces of E .
 2. Show that $F \cap G = \{0_{\mathbb{R}^4}\}$.
 3. Let $(x, y, z, t) \in \mathbb{R}^4$. Determine $a \in \mathbb{R}$ such that $(x - 2a, y + a, z, t - a) \in F$.
 4. Deduce that F and G are complementary subspaces of E ($E = F \oplus G$).

Good luck!